

ShopNest Store Power BI Capstone Project Report

1. Introduction

This report presents the **Power BI dashboard developed for the ShopNest Store Capstone Project**. The dashboard was designed to analyze business performance from multiple perspectives, including product sales, delayed deliveries, payment method usage, product ratings, regional sales distribution, seasonal sales variation, and yearly revenue growth.

The objective of the dashboard is to convert raw transactional data into a clear and interactive analytical view that supports better understanding of sales and operational performance. All required questions from the assessment were addressed through suitable visuals, filters, and dashboard interactions.

2. Dashboard Overview

The dashboard was created in **a single main sheet**, as required in the assessment instructions, and includes all visuals corresponding to the given **8 questions**. In addition to the main dashboard page, a separate **drill through detail page** was created for monthly delivery analysis, allowing more detailed review of delayed and on-time orders.

The dashboard includes:

- KPI cards for high-level business summary
- interactive slicers for filtering the dashboard view
- bar, column, donut, and line charts for comparative analysis
- a **drill through** feature for more detailed operational analysis

The dashboard is organized into analytical sections to improve readability and presentation:

- **Product Analysis**
- **Order Analysis**
- **Customer & Regional Analysis**
- **Time & Revenue Analysis**

Note: The insights below are based on the current slicer selection (**year 2017**) shown in the dashboard.

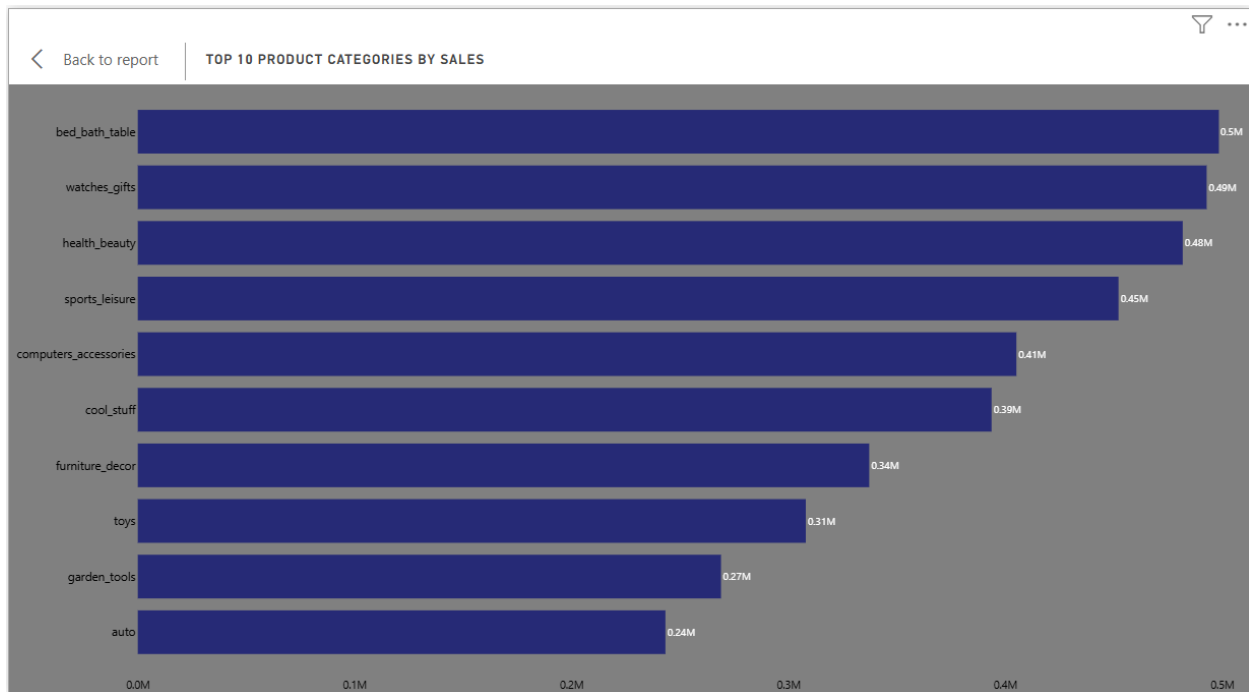
3. Task-wise Sections

3.1 Question 1 — Top Categories by Total Price

Question Statement:

Identify and visually represent the top 10 product categories by total sales.

Visualization:



Explanation:

This visual presents the **top 10 product categories by total sales** using a clustered bar chart. In the current dashboard view, **Health & Beauty** is the highest-performing category followed by **Watches Gifts**. **Bed Bath Table**, **Sports Leisure**, and **Computers Accessories** also contribute strongly, while categories such as **Baby** and **Cool Stuff** appear at the lower end of the top 10 list.

The chart shows that category performance is **not evenly distributed**, as a few categories contribute significantly more sales than others. This indicates that ShopNest Store's sales are more strongly concentrated in selected product segments, with the top categories acting as major revenue drivers in the current view.

Analytical Interpretation:

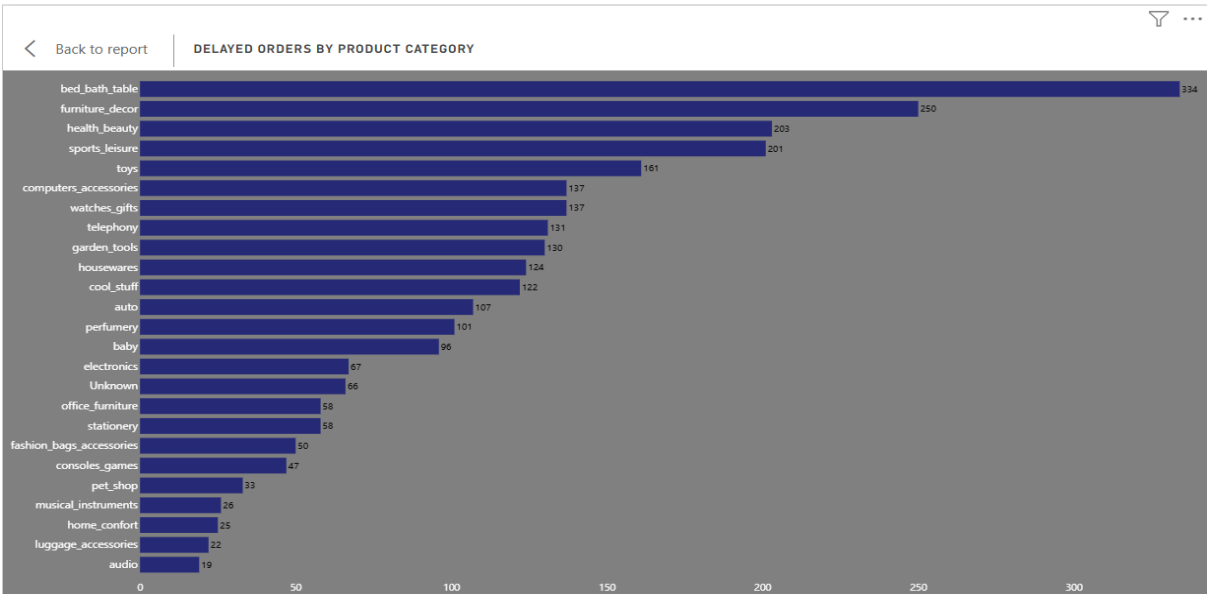
The visual suggests that customer demand is stronger in specific categories rather than spread uniformly across the catalog. The gap between the leading and lower-ranked categories indicates that some product groups have much stronger market traction and sales contribution.

3.2 Question 2 — Delayed Orders Analysis

Question Statement:

Determine the number of delayed orders in each category. An order is considered delayed if the actual delivery date is later than the estimated delivery date.

Visualization:



Explanation:

This visual shows the **number of delayed orders by product category** using a clustered bar chart. In the current dashboard view, **Health & Beauty** has the highest delayed-order count at approximately **571**, followed by **Bed Bath Table** with around **477**. **Sports Leisure**, **Computers Accessories**, and **Watches Gifts** also record relatively high delay counts, while categories toward the bottom of the chart show lower delayed-order volumes.

The chart makes it clear that delayed deliveries are **more concentrated in some categories than others**. Since these categories also tend to be commercially active, the delay pattern may reflect higher order pressure, more complex logistics, or fulfillment inefficiencies affecting certain product groups more heavily.

Analytical Interpretation:

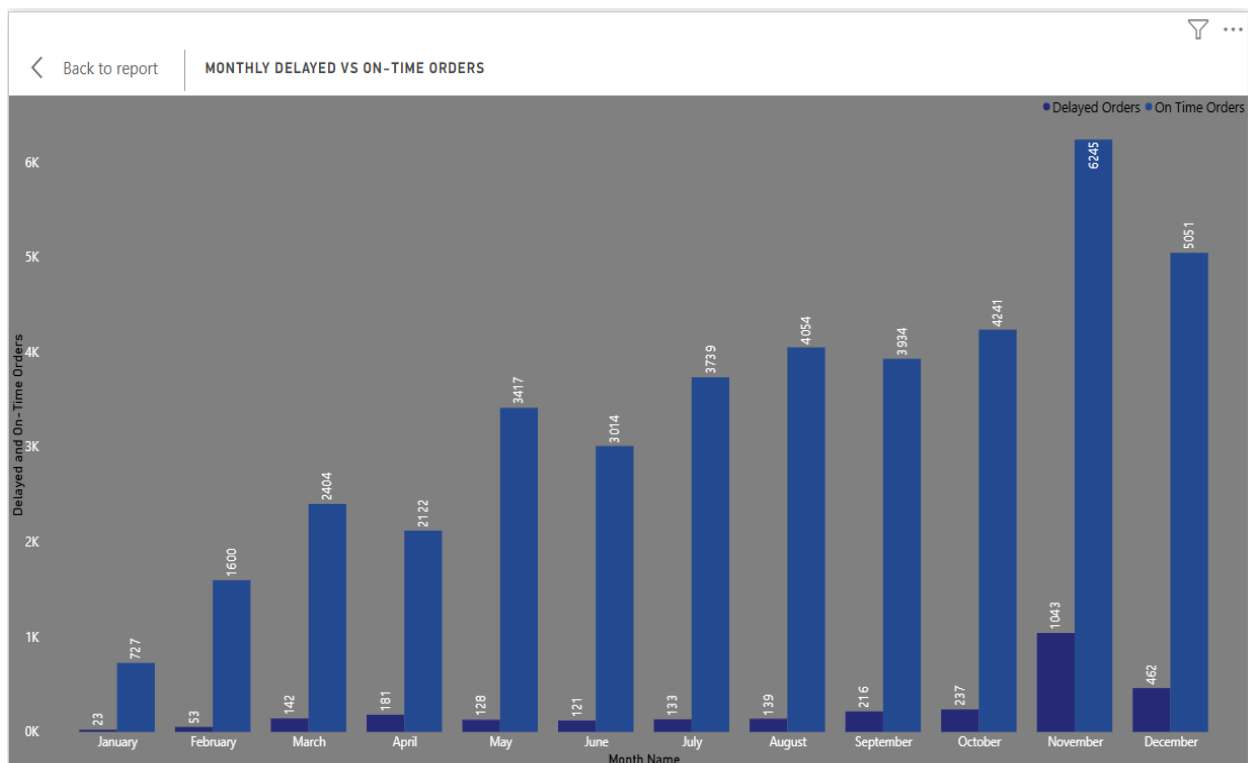
The visual indicates that delivery performance is not consistent across all categories. Some categories appear more vulnerable to operational delays, which may influence customer satisfaction and post-purchase experience. This makes category-level delay tracking important for identifying where fulfillment performance needs closer attention.

3.3 Question 3 — Monthly Comparison of Delayed and On-Time Orders

Question Statement:

Create a dynamic visual that compares the number of delayed orders to the number of orders received earlier for each month. Utilize the drillthrough cross-report feature to provide a detailed analysis of late and on-time deliveries.

Visualization:



According to 2017

Drill through Page



Explanation:

This visual compares **delayed orders and on-time orders month-wise** using a clustered column chart. In the current dashboard view, **on-time orders remain consistently higher than delayed orders across all displayed months**, which indicates that most deliveries are still being completed within the expected timeline.

A drillthrough page was also created to support more detailed order-level analysis for the selected month, making the visual dynamic and more useful for operational review.

Analytical Interpretation:

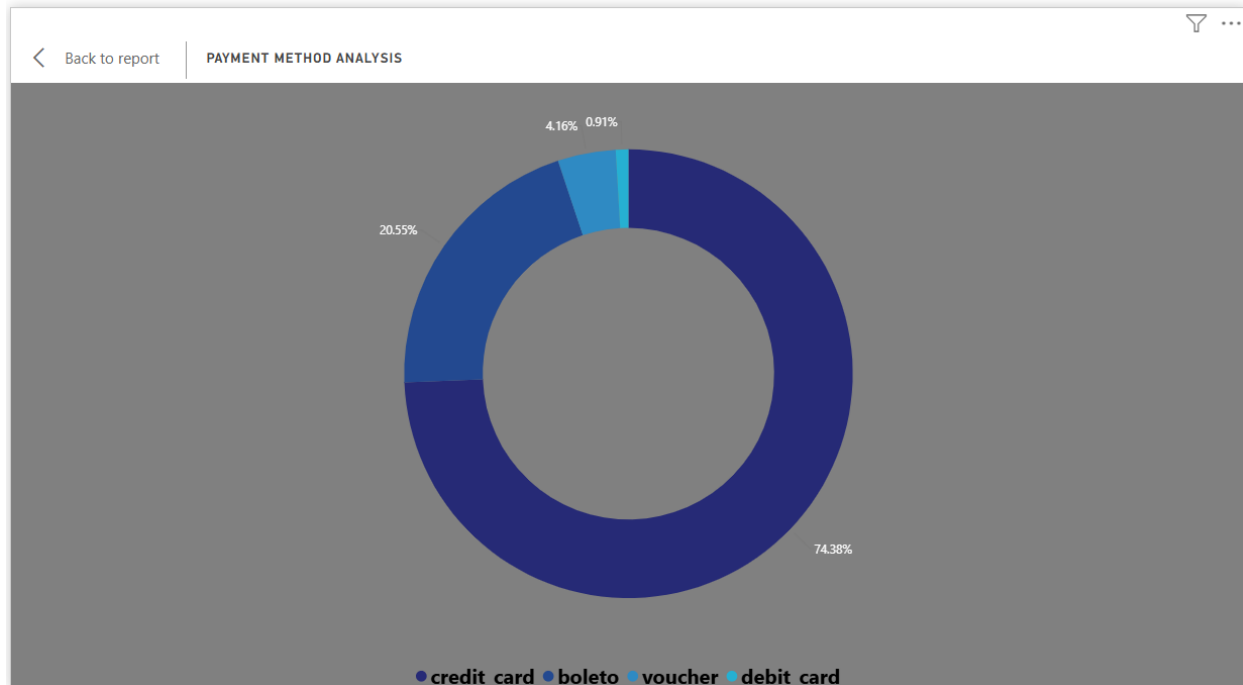
The chart shows that ShopNest Store’s delivery performance is generally stronger on the **on-time** side, but delay levels still fluctuate over time. This suggests that delivery operations are broadly stable, though some months experience higher pressure than others. The monthly comparison helps reveal these variations more clearly than a single summary metric would.

3.4 Question 4 — Payment Method Analysis

Question Statement:

Analyze the most frequently used payment methods by customers using a visually appealing representation, such as a pie chart or other suitable visuals.

Visualization:



Explanation:

This visual presents the **most frequently used payment methods** by customers using a donut chart. In the current dashboard view, **Credit Card** is the dominant payment method. **Boleto** is the second most used method, while **Voucher** and **Debit Card** make up much smaller shares.

The chart clearly shows that payment usage is **heavily concentrated in credit card transactions**, with all other payment methods forming a comparatively smaller portion of customer activity. This makes the customer payment preference pattern easy to understand at a glance.

Analytical Interpretation:

The visual indicates that customers strongly prefer **credit card** over other payment methods in the current view. The gap between credit card and the other options is large, showing that payment behavior is not evenly distributed. This suggests a strong dependence on one payment mode, while the remaining methods function as secondary options.

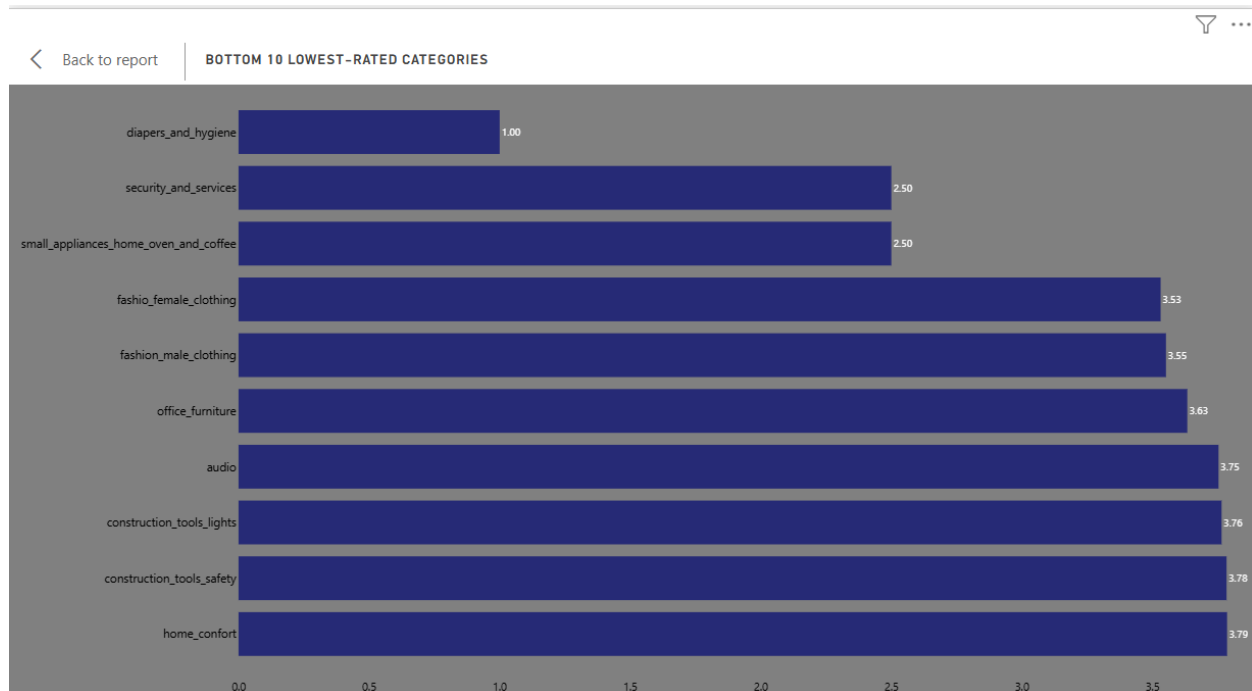
3.5 Question 5 — Product Rating Analysis

Question Statement:

Determine the top 10 highest-rated products and the bottom 10 lowest-rated products using a bar or column chart.

Visualization:





Explanation:

This section presents the **top 10 highest-rated** and **bottom 10 lowest-rated** items using bar charts and average review score as the comparison metric. In the current dashboard, the analysis is shown at the **product category level** instead of the individual product level.

This approach was used for the following reasons:

- The dataset/model does **not provide a clean product-name field** suitable for dashboard presentation.
- Using **product_id** would make the visual highly technical and difficult to interpret.
- Ratings are connected through **order-level review data**, so direct product-wise presentation is not very reader-friendly in the current model.
- Aggregating ratings at the **product category level** provides a clearer and more meaningful comparison.

In the current dashboard view, the highest-rated categories remain clearly above the average rating level, while the lowest-rated categories stay visibly below the top performers. This creates a clear separation between stronger and weaker category groups in terms of customer satisfaction.

Analytical Interpretation:

The chart indicates that customer satisfaction is **not evenly distributed across the catalog**. Some category groups consistently perform better in review scores, while others remain weaker. This suggests that customer experience, product quality, or expectation

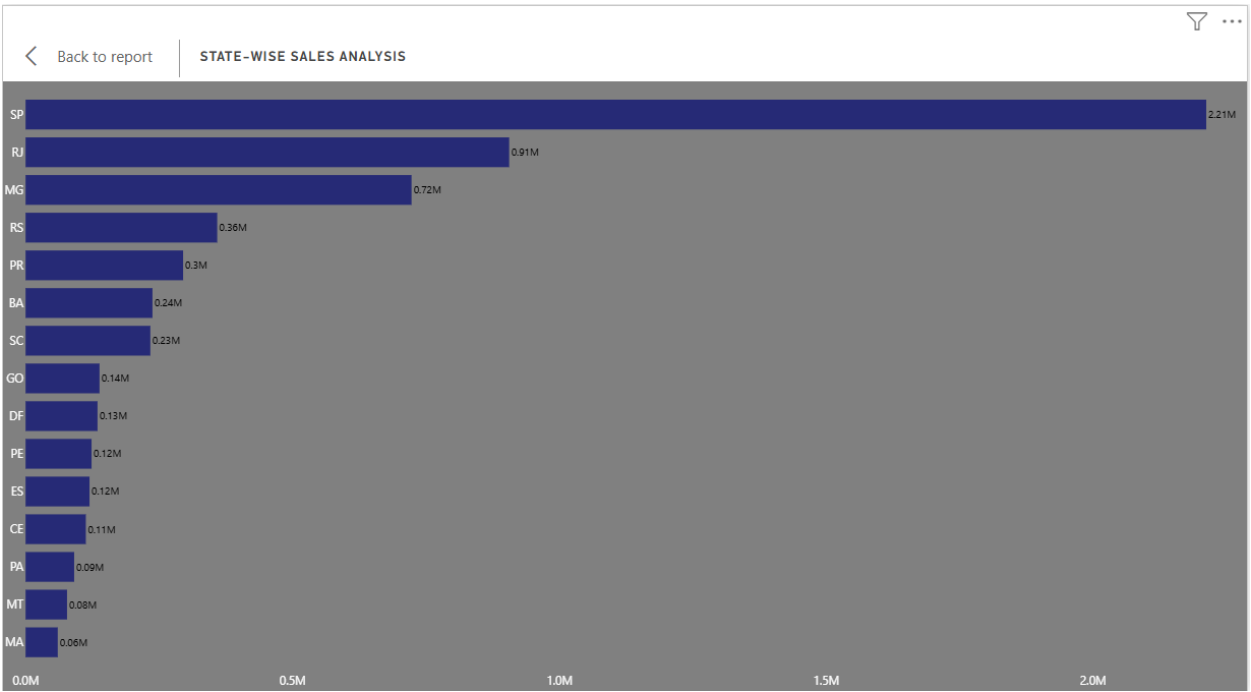
match may be stronger in certain categories than in others. Although the question mentions products, category-level analysis gives a more readable and analytically meaningful interpretation in the present dashboard model.

3.6 Question 6 — State-wise Sales Analysis

Question Statement:

Identify and visually represent states with high and low sales, providing a clear understanding of regional sales performance.

Visualization:



Explanation:

This visual represents **state-wise sales performance** using a clustered bar chart. It compares total sales across states and makes it easy to identify which states contribute more strongly to overall business performance.

In the current dashboard view, **SP** is the strongest-performing state by a very large margin, followed by **RJ** and **MG**. Several other states contribute moderate sales, while the lower end of the chart shows states with much smaller sales contribution. The distribution is therefore highly uneven across regions.

Analytical Interpretation:

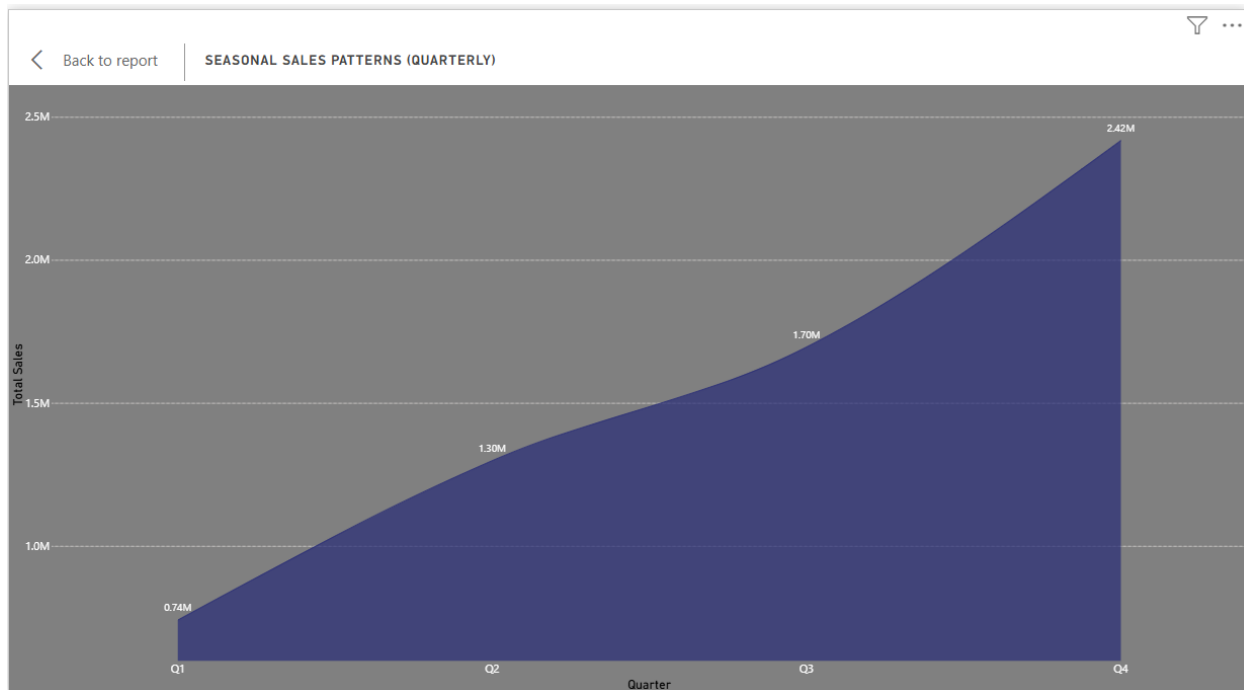
The chart suggests that sales are **geographically concentrated** rather than evenly distributed. A small number of states appear to drive most of the sales value, while other states play a much smaller role. This indicates that regional demand strength is much higher in selected states, making state-level comparison useful for understanding where ShopNest Store performs most strongly.

3.7 Question 7 — Seasonal Sales Patterns

Question Statement:

Investigate and visualize any seasonal patterns (Quarterly) or trends in sales data over the course of the year.

Visualization:



Explanation:

This visual uses a line chart to present **quarterly sales movement** and highlight changes in sales performance across different parts of the year. It helps show whether sales remain stable throughout the year or vary from quarter to quarter.

In the current dashboard view, the line pattern shows clear quarter-wise variation rather than a flat trend. Some quarters perform better than others, indicating that sales intensity

changes during the year. This confirms the presence of a quarterly pattern in sales behavior.

Analytical Interpretation:

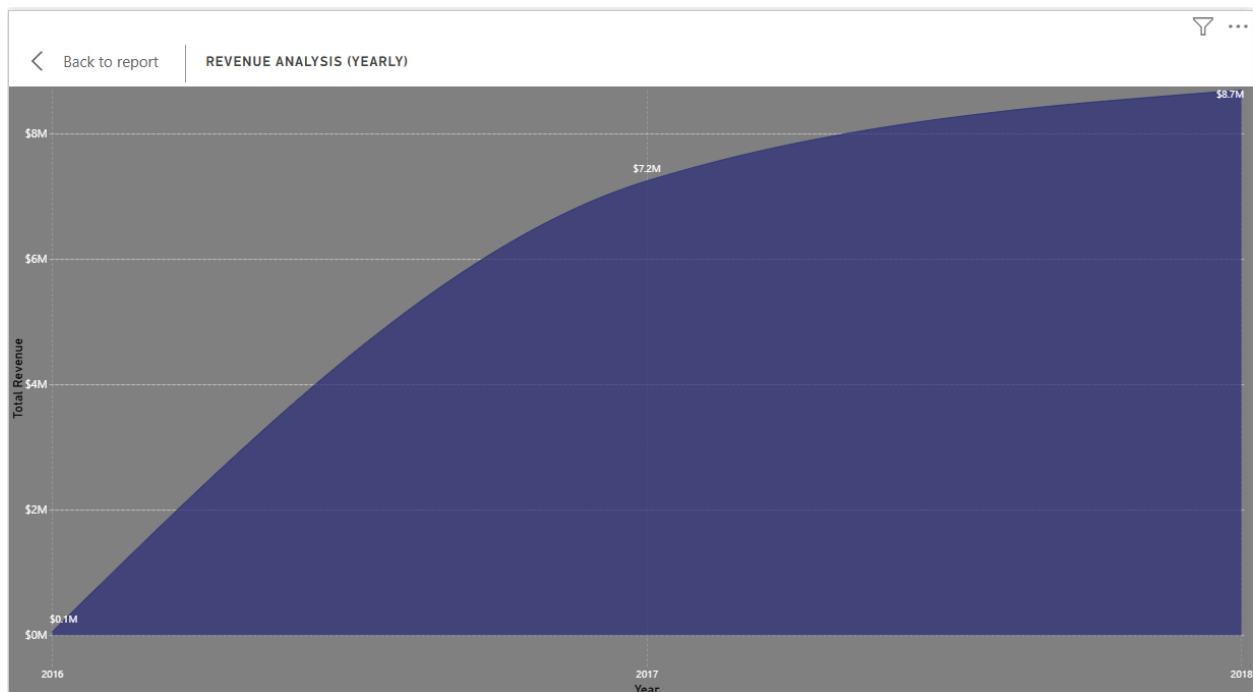
The chart suggests that sales are influenced by **time-based variation**, not distributed uniformly across all quarters. This means that certain periods of the year contribute more strongly to sales performance than others. The quarterly comparison helps reveal this seasonal movement in a more structured way than a single total sales figure.

3.8 Question 8 — Revenue Analysis

Question Statement:

Determine the total revenue generated by ShopNest Store and analyze how it changes over time (Yearly). Represent this information through suitable visuals to highlight trends and patterns.

Visualization:



Explanation:

This visual presents the **yearly revenue trend** using a line chart. It shows how total revenue changes over time and helps identify whether the business is growing, declining, or remaining stable across years.

In the current dashboard, the chart shows a strong upward movement from **2016 to 2018**. Revenue is lowest in 2016, rises sharply in 2017, and continues increasing in 2018. This creates a clear positive trend over time rather than a fluctuating or declining pattern.

Analytical Interpretation:

The chart indicates that ShopNest Store experienced **steady business growth across the observed years**. The revenue movement is clearly upward, suggesting improving commercial performance over time. Among all visuals in the dashboard, this chart gives one of the clearest overall signals of long-term business expansion.

4. Conclusion

The ShopNest Store dashboard provides a comprehensive and interactive view of overall business performance. It identifies the top-performing product categories, highlights delayed delivery patterns, shows customer payment preferences, compares rating performance, explains state-wise sales concentration, reveals quarterly sales variation, and presents the yearly revenue trend.

The dashboard demonstrates that sales and revenue are concentrated in selected categories and regions, delivery performance varies across categories and months, customer payment behavior is strongly centered on credit card usage, and business revenue shows a clear upward trend over time. Overall, the dashboard successfully transforms the dataset into a structured analytical report that is visually clear, interactive, and aligned with the capstone assessment requirements.